ultrahot gas, expanding fast in all directions like the fireball ignited by a nuclear bomb blast or by the explosion of a gas pipeline. Except that the big bang was not destructive (so far as we know). Instead, it *created* everything in our universe, or rather the seeds for everything.

I would love to write a long chapter about the big bang, but with great force of will I restrain myself. We don't need it for the rest of this book.

Galaxies

As our universe expanded, its hot gas cooled. In some regions the gas's density was a bit higher than in others, randomly. When the gas got cold enough, gravity pulled each high-density region inward on itself, giving birth to a galaxy (a huge cluster of stars and their planets and diffuse gas between the stars); see Figure 2.1. The earliest galaxy was born when the universe was a few hundred million years old.

There are roughly a trillion galaxies in the visible universe. The largest galaxies contain a few trillion stars and are about a million light-years across;[3] the smallest, about 10 million stars and a thousand light-years across. At the center of most every large galaxy there is a huge black hole (Chapter 5), one that weighs a million times the sun's weight or more.[4]